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Rhodium-Catalyzed Asymmetric Conjugate Alkynylation/Aldol Cyclization Cascade for the Formation of α-Propargyl-β-hydroxyketones

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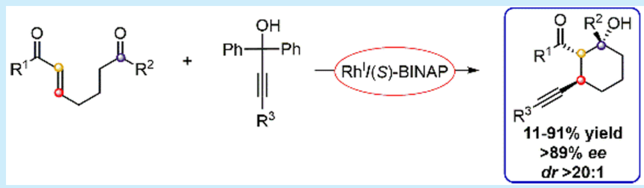
Supporting Information

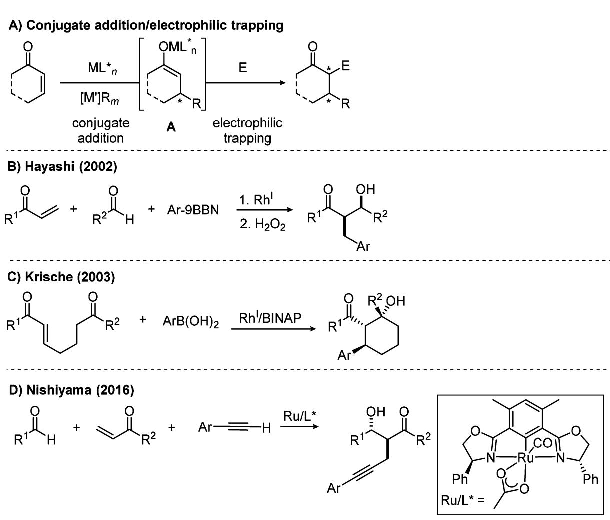
ABSTRACT: A rhodium-catalyzed conjugate alkynylation/aldol cyclization cascade was developed. Densely functionalized cyclic α-propargyl-β-hydroxyketones were synthesized with simultaneous formation of a C(sp)−C(sp^3) bond, a C(sp^3)−C(sp^3) bond, as well as three new contiguous stereocenters. The transformation was achieved with excellent enantio- and diastereoselectivities using BINAP as the ligand. The synthetic utility of the newly installed alkynyl moiety was exhibited by subjecting the products to an array of derivatizations.



Catalytic asymmetric tandem transformations represent an efficient approach to construct complex molecular scaffolds via the sequential formation of multiple bonds and stereocenters, without the need to isolate intermediates.[1] This strategy makes tandem transformations attractive from an economical and environmental standpoint. An example is the transition-metal-catalyzed conjugate addition/electrophilic trapping cascade, which is initiated by a Michael addition. A nucleophilic metal enolate intermediate **A** is generated, which subsequently undergoes electrophilic trapping (Scheme 1A).[2]

Hydride,[3] alkyl,[4] and aryl[5] species have been employed as nucleophilic partners for this class of transformation. Hayashi and Krische concurrently demonstrated the use of a rhodium catalyst with an arylboron reagent for the conjugate arylation/aldol cascade to furnish β-hydroxyketones (Scheme 1B,C).[5a,b] In contrast, a single report by Nishiyama detailed the use of alkynyl nucleophiles as the initiator under ruthenium catalysis, affording α-propargyl-β-hydroxyketones with moderate enantio- and diastereoselectivity (Scheme 1D).[6]

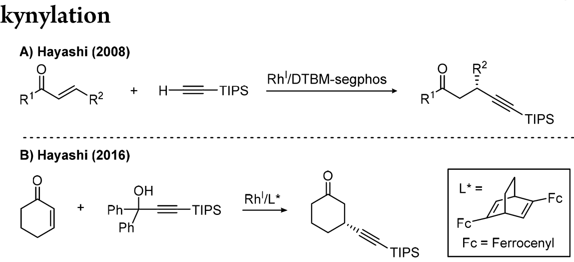
Scheme 1. Conjugate Addition/Electrophilic Trapping Cascade



Alkynes are versatile synthetic building blocks capable of undergoing a wide array of manipulations.[7] Over the past decade, extensive effort has been devoted to developing transition-metal-catalyzed methods to introduce an alkyne moiety enantioselectively to prochiral carbonyls through the formation of C(sp)—C(sp^3) bonds.[8,9] Rhodium catalysis has been recently reported to enable the enantioselective conjugate alkynylation of α,β-unsaturated carbonyls.[9] Seminal work by Hayashi showed that judicious selection of chiral, bulky bisphosphine ligands was key to mitigate the propensity for alkyne homodimerization (Scheme 2A).[9b] Recent developments in this area culminated with the use of easily accessible, bulky alkynyl(diphenyl)carbinol as the alkynylating reagent to achieve the desired transformation (Scheme 2B).[9e]

We envisioned that rhodium could function as a suitable catalyst for the conjugate alkynylation/aldol cascade for the

Scheme 2. Rhodium-Catalyzed Enantioselective Conjugate Alkynylation



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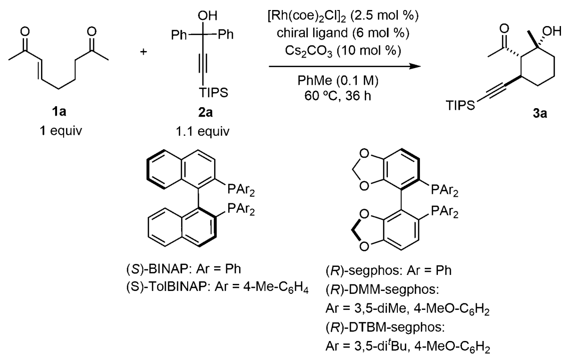
simultaneous formation of a C(sp)—C(sp^3) bond and a C(sp^3)—C(sp^3) bond. Applying literature conditions for the rhodium-catalyzed enantioselective alkynylations[9b,f] to the model substrate **1a** and alkyne **2a** in the presence of chiral bisphosphine ligands furnished the desired α-propargyl-β-hydroxyketones **3a** with three contiguous stereocenters (Table 1). No significant effect on yield or enantioselectivity

Table 1. Chiral Ligand Screening[a]



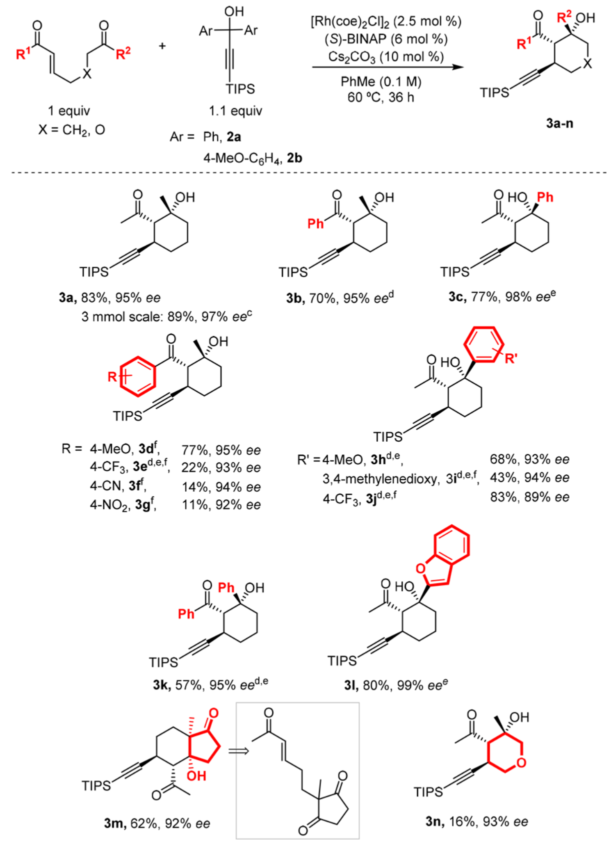
entry	ligand	yield[b] (%)	dr[c]	ee[d] (%)
1	**(*S*)-BINAP**	**>99 [83]**	**>20:1**	**95**
2	(*S*)-TolBINAP	96 [83]	>20:1	94
3	(*R*)-segphos	>99 [83]	>20:1	−95
4	(*R*)-DMM-segphos	90 [78]	>20:1	−95
5	(*R*)-DTBM-segphos	>99 [87]	>20:1	−95

[a]Reactions conducted on 0.2 mmol scale. [b]Yields determined by ^{1}H NMR spectroscopic analysis of the crude reaction mixture using 1,3,5-trimethylbenzene as the internal standard. Isolated yields are reported in square brackets. [c]Determined by ^{1}H NMR spectroscopic analysis of the crude reaction mixture. [d]ee values are determined by HPLC analysis on a chiral stationary phase. nd = not determined. coe = *cis*-cyclooctene.

was observed when variants of the BINAP and segphos classes of ligands were examined (entries 1–5). A solvent screen indicated toluene as the optimal solvent.[10] Furthermore, a high degree of diastereocontrol was observed in all cases.

With BINAP as the optimal ligand, the keto-enone substrate scope was explored (Scheme 3). Substrates bearing a methyl or phenyl substituent on the enone afforded the desired products in high yields and enantioselectivities (**3a**,**b**). The reaction was amenable to scale up (3 mmol) to give **3a** with a comparable yield and enantioselectivity of 89% and 97% ee, respectively. Product **3c** containing a highly crowded tertiary alcohol was furnished with good yield. Diminished reactivity was observed when the enone moiety bears electron-deficient aryl rings (**3e**–**g**), which was consistent with the decrease in nucleophilicity of the enolate intermediate that was formed. Conversely, electron-rich aryl rings on the ketone moiety decreased the yield due to lower electrophilicity (**3h**,**i**). Phenyl groups on both the enone and the ketone moieties exhibited slightly diminished reactivity (**3k**). Products containing a benzofuran moiety (**3l**) or 5,6-fused rings with four contiguous stereocenters (**3m**) were furnished successfully. In addition, a trisubstituted tetrahydropyran (**3n**) was obtained, albeit with low yield. However, substrates bearing an acrylate moiety or an ester secondary electrophile did not produce any cyclized product. Similarly, cyclization did not occur to form either five or seven-membered rings when the tether length of the keto-enone substrate was varied.[10] Regardless, the reaction displayed an enantioselectivity

Scheme 3. Keto-enone Substrate Scope[a,b]



[a]All dr > 20:1. [b]See the Supporting Information for unreactive substrates. [c]With [Rh(cod)Cl]$_2$. [d]72 h reaction time. [e]With **2b**. [f]With [Rh(C_2H_4)$_2$Cl]$_2$.

of 89–99% ee and diastereoselectivity of >20:1 for all reactive substrates tested.[11]

While investigating the alkyne substrate scope, we observed that the reactivity of the transformation was contingent upon having sufficient steric bulk flanking the propargylic carbon (Scheme 4). Due to extensive homo-dimerization of the alkyne, the reaction completely halted when unsubstituted or monosubstituted aryl was present on the alkyne. Nevertheless, reactivity was restored when 2,6-dimethylphenyl acetylene was used (**3o**). Linear alkyls, cyclohexyl, as well as heterocycles such as tetrahydropyran and Boc-protected piperidine were also coupled in this fashion (**3o**–**t**). Furthermore, a biologically relevant moiety containing multiple existing stereocenters such as the dihydrocholesterol scaffold can be coupled efficiently with excellent diastereoselectivity (**3u**,**v**). Finally, polycyclic products can be furnished via this transformation with high enantioselectivities and modest yields (**3w**,**x**).

To demonstrate the synthetic versatility of the alkyne functional unit, derivatization of α-propargyl-β-hydroxyketone **3a** was studied (Scheme 5). An alternative method to synthesize products bearing a phenyl acetylene moiety can be accomplished using a one-pot deprotection/Sonogashira coupling process. A 4-bromobenzene ring was appended to the alkyne to generate **4**. The absolute stereochemistry of **3a** was unambiguously established by single X-ray crystallography of *ent*-**4**, and the stereochemistry of the other products was assigned by analogy.[12] Deprotection of the triisopropylsilyl group in the presence of CuI afforded the terminal alkyne **5**,

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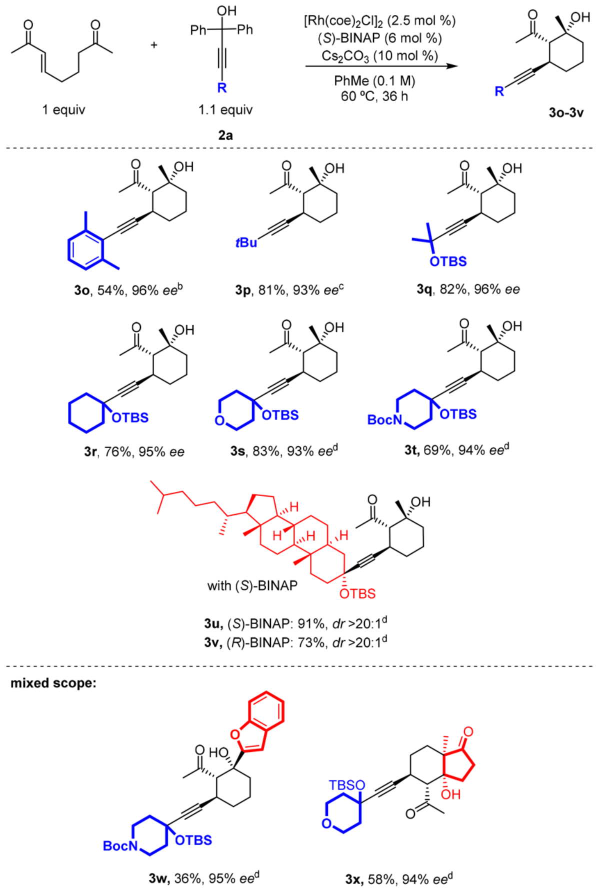
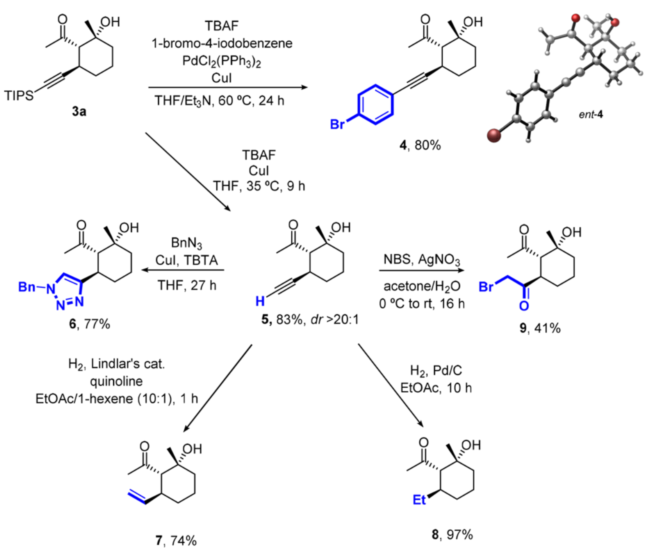
Scheme 4. Alkyne Substrate Scope[a]



[a]All dr > 20:1. [b]With 1.5 equiv of alkyne. [c]With 1.3 equiv of alkyne. [d]72 h reaction time.

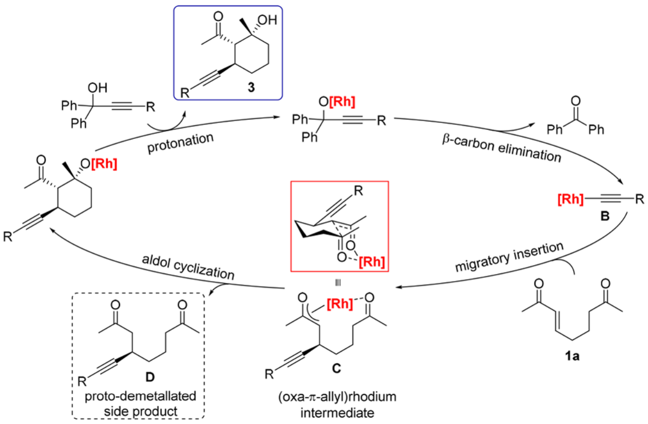
Scheme 5. Product Derivatizations



which could be further functionalized. It was presumed that CuI prevents epimerization by chelating the oxygen atoms. Terminal alkyne **5** underwent copper-catalyzed 1,3-dipolar cycloaddition smoothly with benzyl azide to form benzyl triazole **6**. Additionally, partial hydrogenation with Lindlar's catalyst and quinoline afforded alkene **7**, while complete hydrogenation yielded **8**. Lastly, silver-mediated halo-hydration produced α-bromoketone **9** at moderate yield.

Based on detailed mechanistic studies on rhodium-catalyzed conjugate additions reported by Hayashi[13] and the precedence established by Krische on rhodium-catalyzed conjugate arylation/aldol cyclization,[5c] a plausible mechanism for the conjugate alkynylation/aldol cyclization was proposed (Scheme 6). The catalytic cycle starts with the coordination of the

Scheme 6. Proposed Mechanism



rhodium catalyst to the propargyl alkoxide followed by β-carbon elimination to generate rhodium−alkynyl species **B** while extruding benzophenone. Alkynylrhodation of enone **1a** forms (oxa-π-allyl)rhodium intermediate **C**. Chelation of the oxygen atoms by rhodium in a favorable six-membered Zimmerman−Traxler-type transition state dictates the *syn*-relationship of the two groups.[14] Finally, cyclization followed by protonation provides the desired product **3**. Depending on the substrate, proto-demetalation of **C** occurred to give side product **D** at 8−20% yields.

In summary, an asymmetric rhodium-catalyzed conjugate alkynylation/aldol cyclization cascade was developed to synthesize highly functionalized cyclic α-propargyl-β-hydroxyketones with multiple contiguous stereocenters. Excellent enantio- and diastereoselectivities of 89−99% ee and dr > 20:1 were observed. The appended alkyne functional group exhibited potential for further useful synthetic manipulations. This reaction also demonstrated the possible use of alkynylrhodation as the initiating step for other widely studied rhodium-catalyzed cascades.[5g,15]

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.8b00153.

Experimental procedures, optimization, characterization, and X-ray data (PDF)

Accession Codes

CCDC 1590594 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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(10) See the Supporting Information.

(11) Alkyne **2b** was used in cases where the product co-elutes with the benzophenone byproduct during flash column chromatography.

(12) *ent*-**4** was synthesized from *ent*-**3a**, which was synthesized during the optimization process when (*R*)-segphos was used as the chiral ligand.

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